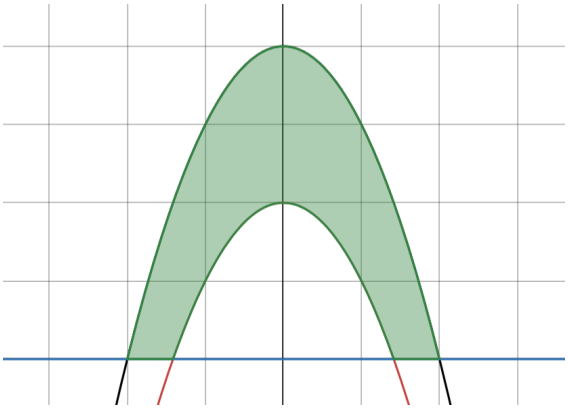
Problem. Compute the area of the plane region depicted below, bounded by the following curves:

$$y = 4 - x^2$$

$$y = 2 - x^2$$

$$y = 0$$



Solution. One way to find the area here is to take the area under the larger parabola and subtract the area under the smaller parabola. The first of these is

$$\int_{-2}^{2} [4 - x^2] \mathrm{d}x = 2 \int_{0}^{2} [4 - x^2] \mathrm{d}x = 2 \left[4x - \frac{x^3}{3} \right] \Big|_0^2 = 2(8 - 8/3) = \frac{32}{3}$$

while the second is

$$\int_{-\sqrt{2}}^{\sqrt{2}} [2 - x^2] \mathrm{d}x = 2 \int_{0}^{\sqrt{2}} [2 - x^2] \mathrm{d}x = 2 \left[2x - x^3/3 \right] \Big|_0^{\sqrt{2}} = 2(2\sqrt{2} - 2\sqrt{2}/3) = \frac{4\sqrt{2}}{3}$$

and so the area of the indicated region is

$$\frac{32 - 4\sqrt{2}}{3}$$

Problem. Consider plane region below the graph of $x^{-1/2}$ over the interval $[1, 4]$. Compute the volume of the solid obtained by revolving this region about the x axis.

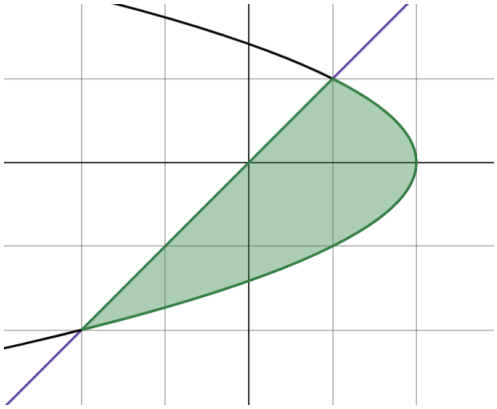
Solution. We can express the volume as an integral in x. The relevant cross section is a disk of radius $x^{-1/2}$, which thus has area $A(x) = \pi x^{-1}$.

$$V = \int_{1}^{4} \pi x^{-1} \mathrm{d}x = \pi \ln(4)$$

Problem. Compute the area of the plane region depicted below, bounded by the following curves:

$$y = x$$

$$x = 2 - y^2$$



Solution. We can express this area as an integral in y. The intersection points are solutions to $y = 2 - y^2 \rightsquigarrow y^2 + y - 2 = 0 \rightsquigarrow (y+2)(y-1) = 0$, so $y = -2$, $y = 1$. The cross sectional width at y is $2 - y^2 - y$, and so the area is

$$A = \int_{-2}^{1} [2 - y^2 - y] \mathrm{d}y = \left[2y - \frac{y^3}{3} - \frac{y^2}{2} \right] \Bigg|_{-2}^{1} = (2 - 1/3 - 1/2) - (-4 + 8/3 - 2) = 9/2$$

Problem. Compute the volume of the solid obtained by revolving the plane region between the curves $y = 2x^2$ and $y = x + 1$ over the interval $[0, 1]$ around the y-axis.

Solution. We can evaluate the volume using the shell method. For this, we need to know the cross sectional height. On the interval $[0, 1]$, the curve $y = x + 1$ is above $y = 2x^2$, so the height is $x + 1 - 2x^2$. The area of a horizontal shell at radius x is then $2\pi x(x + 1 - 2x^2) = 2\pi(x^2 + x - 2x^3)$, and so the volume is

$$V = 2\pi \int_{0}^{1} [x^2 + x - 2x^3] \mathrm{d}x = \left[\frac{x^3}{3} + \frac{x^2}{2} - \frac{x^4}{2} \right] \Bigg|_{0}^{1} = \frac{2\pi}{3}$$

Problem. Evaluate the following integral.

$$\int_{-\pi/4}^{\pi/4} \frac{\cos(2x)}{2 + \sin(2x)} \mathrm{d}x$$

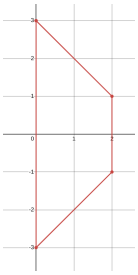
Solution. We first make the substitution $u = 2x$, transforming the integral into

$$\frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{\cos(u)}{2 + \sin(u)} \mathrm{d}u$$

We then make the substitution $v = 2 + \sin(u)$, transforming it into

$$\frac{1}{2}\int_1^3 \frac{1}{v}\mathrm{d}v = \frac{\ln(3)}{2}$$

Problem. Find the volume of a solid whose base is the trapezoid depicted below, and whose cross sections perpendicular to the x-axis are triangles whose base and height are equal.



Solution. The base of each triangular cross section is the vertical line segment through the trapezoid at that x value, which has length $(3 - x) - (-3 + x) = 6 - 2x$. Therefore, the area $A(x)$ of the cross section at x is

$$\frac{(6-2x)^2}{2}$$

and the volume is

$$V = \int_0^2 \frac{(6-2x)^2}{2}\mathrm{d}x$$

Using $u = 6 - 2x$, this becomes

$$-\frac{1}{2}\int_6^2 \frac{u^2}{2}\mathrm{d}u = \frac{1}{4}\int_2^6 u^2\mathrm{d}u = \frac{1}{4}\left[\frac{u^3}{3}\right]\Big|_2^4 = \frac{64-8}{12} = \frac{56}{12} = \frac{14}{3}$$

Problem. Find the average value of $2\sin(x) + \cos(x)$ on the interval $[3\pi, 9\pi/2]$.

Solution. This average is

$$\frac{1}{9\pi/2 - 3\pi}\int_{9\pi/2}^{3\pi} [2\sin(x) + \cos(x)]\mathrm{d}x$$

Using the periodicity of the function $2\sin(x) + \cos(x)$, this is

$$\frac{2}{3\pi}\int_{-\pi}^{\pi/2} [2\sin(x) + \cos(x)]\mathrm{d}x = \frac{2}{3\pi}\left[-2\cos(x) + \sin(x)\right]\Big|_{-\pi}^{\pi/2} = \frac{2}{3\pi}(1-2) = -\frac{2}{3\pi}$$

Problem. Consider the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 3x + 1$.

(a) Explain why this function is injective.

(b) Explain why this function is surjective.

(c) Let g be the inverse of f. Using $f(-1) = -3$, compute $g'(-3)$.

Solution. (a) The function f has derivative $f'(x) = 3x^2 + 3$, which is positive everywhere. Therefore, f is strictly increasing, and so is injective.

(b) The function f is an odd degree polynomial whose leading coefficient is positive, so we have

$$\lim_{x \to +\infty} f(x) = +\infty$$

$$\lim_{x \to -\infty} f(x) = -\infty$$

This means that f is unbounded in both directions, so because it is continuous, the intermediate value theorem implies that it is surjective.

(c) The general formula for the derivative of an inverse tells us that

$$g'(-3) = \frac{1}{f'(g(-3))}$$

We have $g(-3) = -1$, and $f'(-1) = 6$, so this comes out to $1/6$.